Tilak Sharma

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Experience

Cofounder

FluidZero AI (Escape Velocity Labs, Inc.) *fluidzero.ai*

- Cofounded a bootstrapped AI document intelligence company for construction (AEC) and compliance; own the extraction and ML stack end to end. Landed a **paying pilot** with a road construction firm plus active construction design partners; placed **2nd at Antler AI Builders Day**.
- Built the retrieval and knowledge layer: ColPali/MUVERA multi vector embeddings with BM25 hybrid search (RRF fusion), and a Neo4j property graph with typed relations (references, conflicts, supersedes) and bitemporal conflict resolution, exposed to downstream agents through a tool layer of six orthogonal query axes.
- Built the parsing pipeline (DocLayout-YOLO, RT-DETR, PaddleOCR) with seam resolution for elements split across pages and per document self calibration of detection thresholds.
- Published a knowledge graph vs. markdown retrieval benchmark (fluidzero.ai/research).

AI Engineer July 2025 – Present

Everest Insurance & Reinsurance *Warren, NJ*

- Lead the reinsurance contract wording intelligence initiative end to end, coordinating five engineers: custom agents (LangGraph, tool use over raw documents) for clause classification and contract translation, running on an execution layer of sandboxed containers on Kubernetes so every agent run is **isolated and reproducible**.
- Building an internal agent framework that generalizes tool calling into a reusable library for engineers across teams.
- Contributed core components of the automated submission intake system, now running **4–5K transactions/month at 85% straight through processing** and cutting average handling time from **1.5–2 hours** of manual BPO review to **15 minutes**; packaged as a common library adopted across business segments.

Machine Learning Engineer May 2024 – March 2025

Mbodi AI *Manhattan, NY*

- Implemented RT-1 and RT-2 robotic transformer architectures for embodied action prediction, working through tokenization of continuous actions and multimodal conditioning of language and vision inputs.
- Extended YOLO detection pipelines (with co-training) to 6D pose estimation on the BOP dataset, parameterizing rotation and translation with custom losses and metrics.
- Developed an ensemble few-shot object detection system combining CV2 features and deep embeddings with YOLO and SAM2 for localization, cutting pipeline latency by **65%** without sacrificing performance.

AI/ML Research Intern July 2024 – September 2024

Artisse AI *Remote, United States*

- Engineered a virtual try-on architecture using **Stable Diffusion**, CLIP, T5, **IP adapter**, and **ControlNet**; optimized training with distributed data parallel, DeepSpeed, and pipeline parallelism. Fine-tuned SAM for automatic masking and dataset labeling.

Software Development Engineer I → II July 2020 – March 2022

Practo *Bangalore, India*

- Revamped the Payments SDK (**+3.5%** payment success rate) and built an in-app insurance point of sale system (**+6%** consultation plan upsell).

Research Projects

Egress Review Gym – RL environment for drawing review 2026

- Building an RL environment and a dataset of construction drawings with planted egress violations to train a small vision model to review code compliance; currently diagnosing why initial runs failed to learn and iterating on how violations are planted.

fluidfield – site walk verification against BIM 2026

- Built an iOS capture app that records ARKit poses, LiDAR depth, RoomPlan geometry, and the world map from a site walk, and a pipeline that registers each walk to the BIM model and to earlier walks (floor height match, FFT search over the floor plan, trimmed ICP).
- Each expected building element gets a verdict of **seen present**, **seen absent**, or **not seen**, so the system abstains instead of guessing when coverage or pose quality is insufficient.

Publications

[1] **T. Sharma H K**, A. R. Kaushik, N. Ratha. "PixMus: Video and Text Conditioned Background Music Generation Using Latent Diffusion." *IEEE Conference on Artificial Intelligence (CAI)*, 2025. *(Master's thesis)*

[2] A. Malik, N. Ratha, B. Yalavarthi, **T. Sharma**, A. Kaushik, C. Jutla. "Confidential and Protected Disease Classifier Using Fully Homomorphic Encryption." *IEEE CAI*, 2024.

[3] **T. Sharma**, N. Ratha, C. Jutla. "Privacy-Preserving Ensemble Learning Using Fully Homomorphic Encryption." *International Conference on Pattern Recognition (ICPR)*, 2024.

[4] B. Yalavarthi, A. R. Kaushik, **T. Sharma**, C. Jutla, N. Ratha. "Secure Sleep Apnea Detection with FHE and Deep Learning on ECG Signals." *ICPR*, 2024.

[5] T. P. Agrawal, B. Yalavarthi, A. R. Kaushik, **T. Sharma**, C. Jutla, N. Ratha. "Privacy-Enhanced Graph Edge Convolutional Networks." *IEEE Western New York Image and Signal Processing Workshop (WNYISPW)*, 2024.

[6] **T. Sharma**, M. Wason, V. Boddeti, A. Ross, N. Ratha. "Fully Homomorphic Encryption Operators for Score and Decision Fusion in Biometric Identification." *IEEE International Workshop on Information Forensics and Security (WIFS)*, 2023.

[7] **H. K. Srinidhi**, H. S. Shreenidhi, G. S. Vishnu. "Smart Hydroponics System Integrating with IoT and Machine Learning Algorithm." *International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT)*, 2020. *(published under former name)*

Education

State University of New York (SUNY) — Buffalo, NY
Master of Science in Engineering Sciences – Artificial Intelligence & Robotics. — *2022 – 2024*

Jain University - School of Technology & Engineering — Bangalore, India
Bachelor of Technology in Computer Science and Engineering. — *2016 – 2020*

Technical Skills

ML: PyTorch, HuggingFace Transformers, DeepSpeed, distributed training, diffusion models, VLMs, RL environments (OpenAI Gym), LangGraph, Anthropic SDK.
Systems: Python, Java, C/C++, Kubernetes, Docker, Airflow, Kafka, Redis, Celery, Neo4j, PostgreSQL, Elasticsearch, FastAPI, MLflow, AWS, Azure.